

CHAPTER 5:

DIFFUSION IN SOLIDS

ISSUES TO ADDRESS...

- How does diffusion occur?
- Why is it an important part of processing?
- How can the rate of diffusion be predicted for some simple cases?
- How does diffusion depend on structure and temperature?

Diffusion

Diffusion - Mass transport by atomic motion

Mechanisms

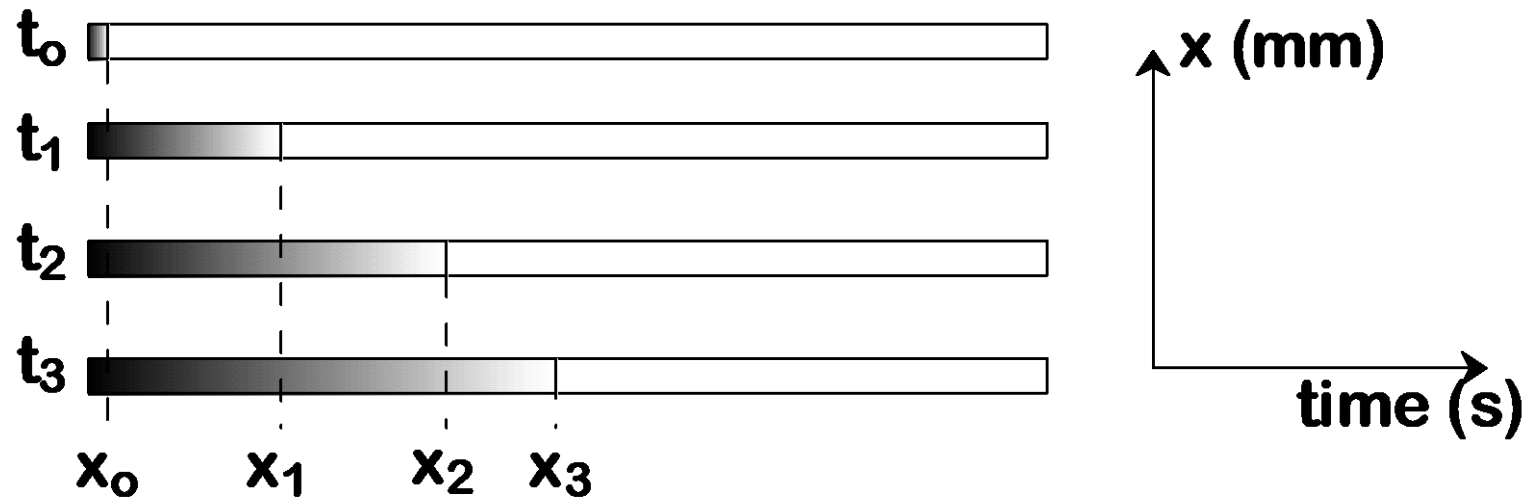
- **Gases & Liquids – random (Brownian) motion**
- **Solids – vacancy diffusion or interstitial diffusion**

Why Study Diffusion ?

- Diffusion plays a crucial role in...
 - Alloying metals => bronze, silver, gold
 - Strengthening and heat treatment processes
 - Hardening the surfaces of steel
 - High temperature mechanical behavior
 - Phase transformations
 - Mass transport during FCC to BCC
 - Environmental degradation
 - Corrosion, etc.

DIFFUSION DEMO

- Glass tube filled with water.
- At time $t = 0$, add some drops of ink to one end of the tube.
- Measure the diffusion distance, x , over some time.
- Compare the results with theory.



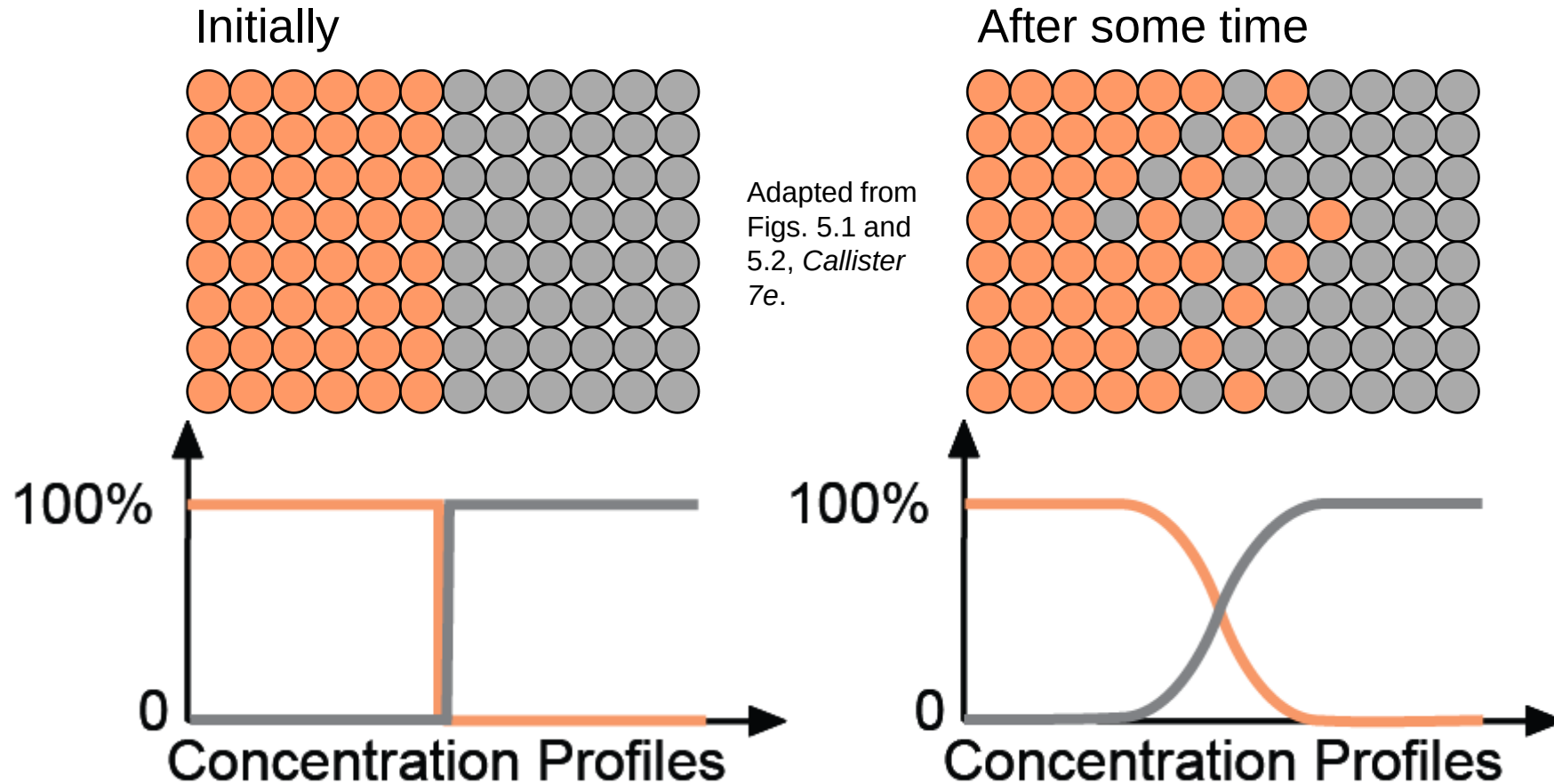
How do atoms move in Solids ?

Why do atoms move in Solids ?

- Diffusion, simply, is atoms moving from one lattice site to another in a stepwise manner
 - Transport of material by moving atoms
- Two conditions are to be met:
 - An empty adjacent site
 - Enough energy to break bonds and cause lattice distortions during displacement
- What is the energy source ?
 - HEAT !
- What else ?
 - Concentration gradient !

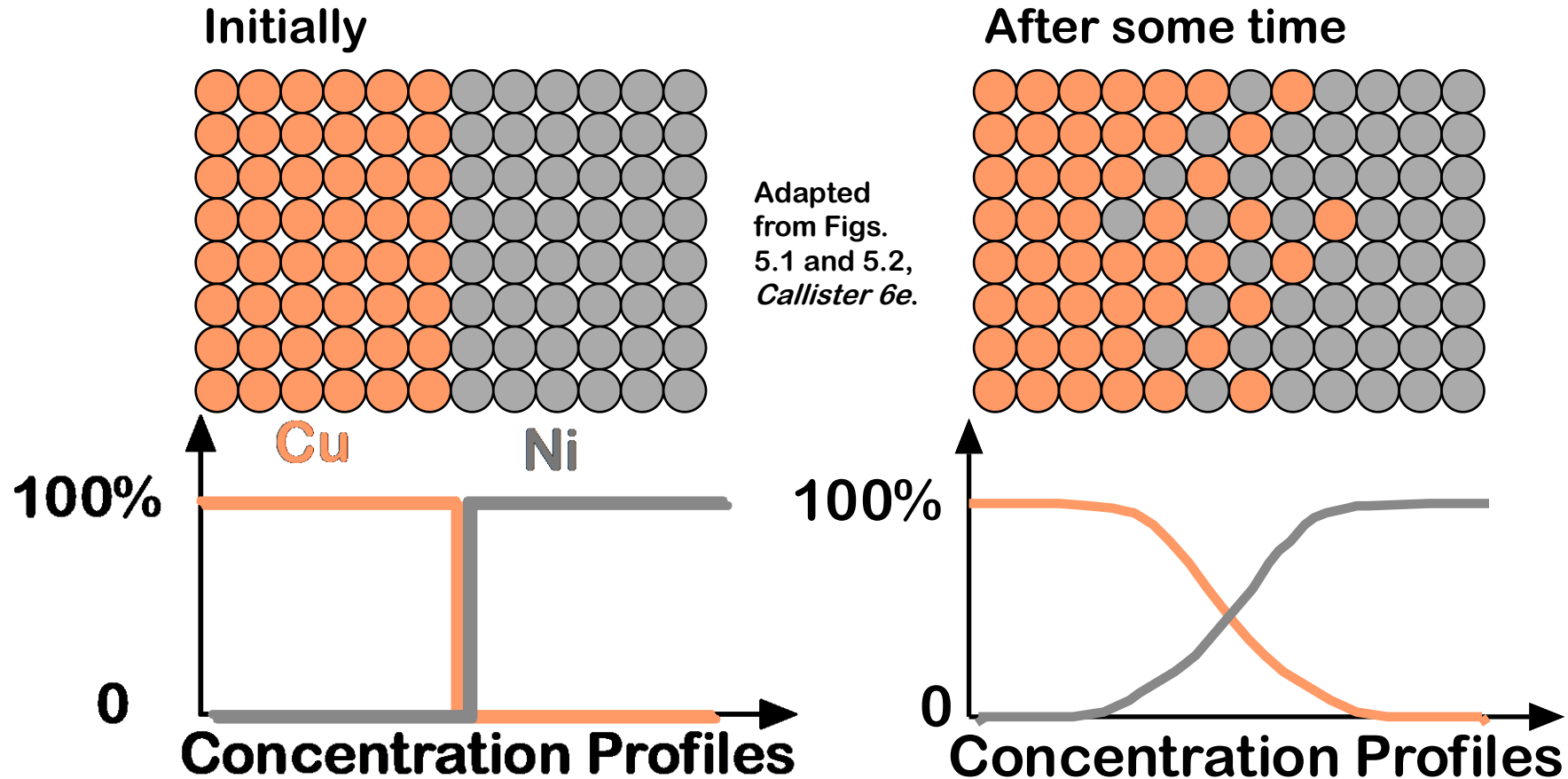
Diffusion

- **Interdiffusion:** In an alloy, atoms tend to migrate from regions of high conc. to regions of low conc.



DIFFUSION: THE PHENOMENA (1)

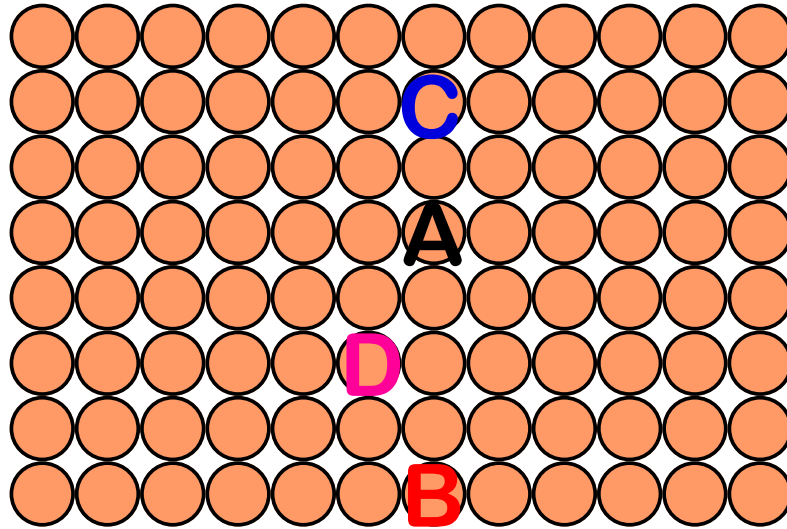
- **Interdiffusion:** In an alloy, atoms tend to migrate from regions of large concentration.



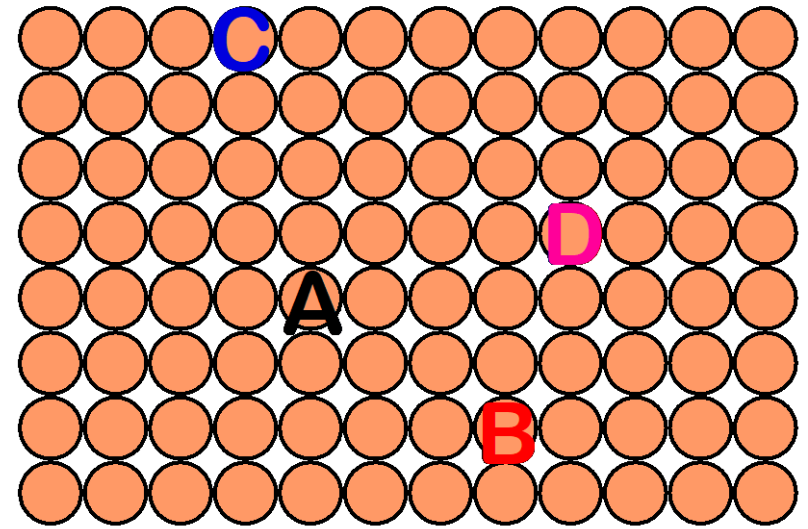
DIFFUSION: THE PHENOMENA (2)

- **Self-diffusion:** In an elemental solid, atoms also migrate.

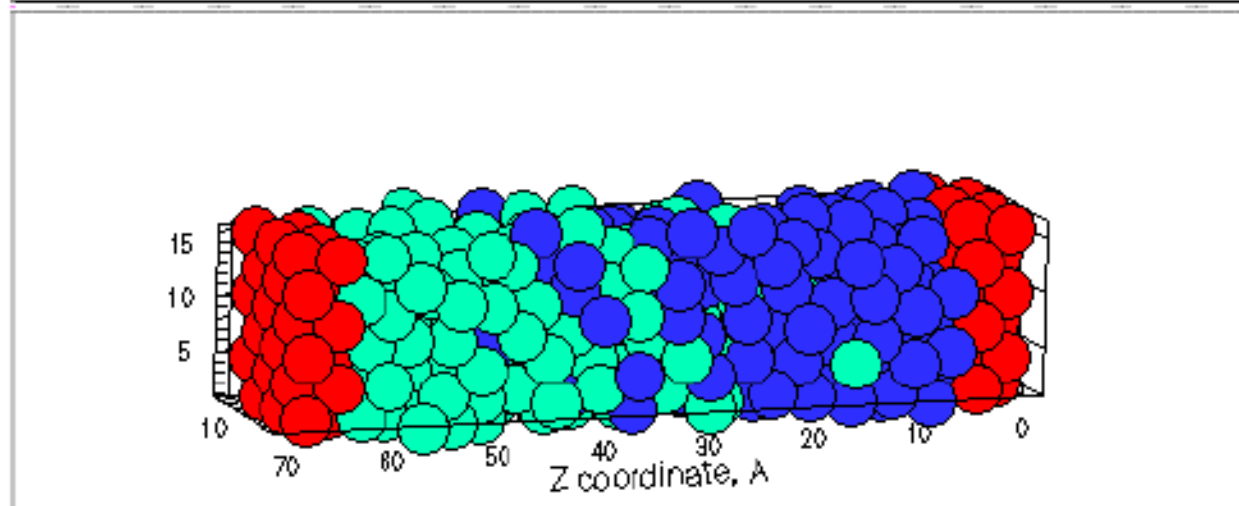
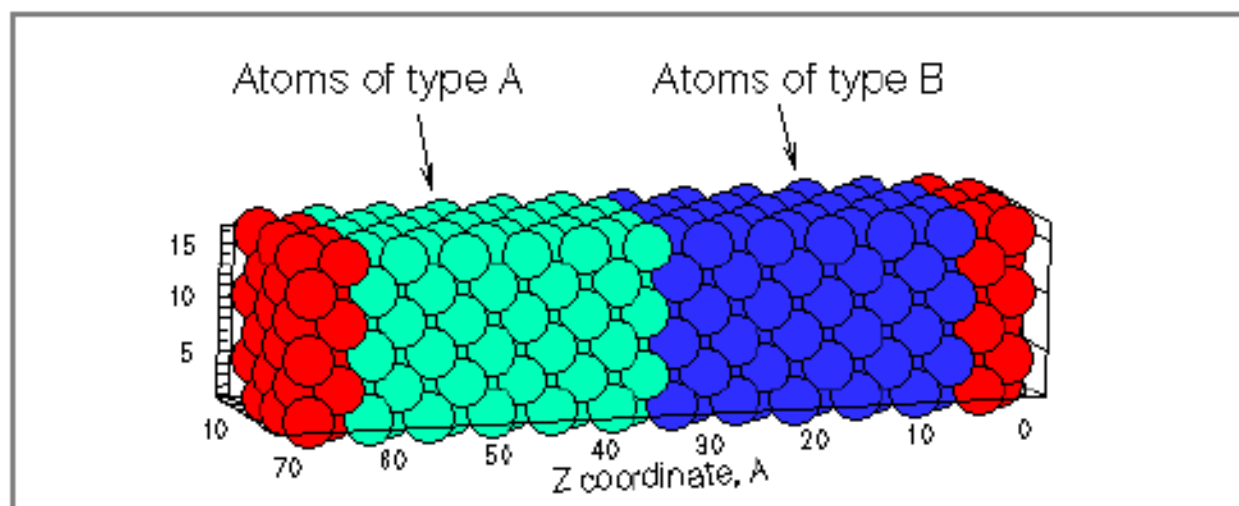
Label some atoms (use isotopes)



After some time

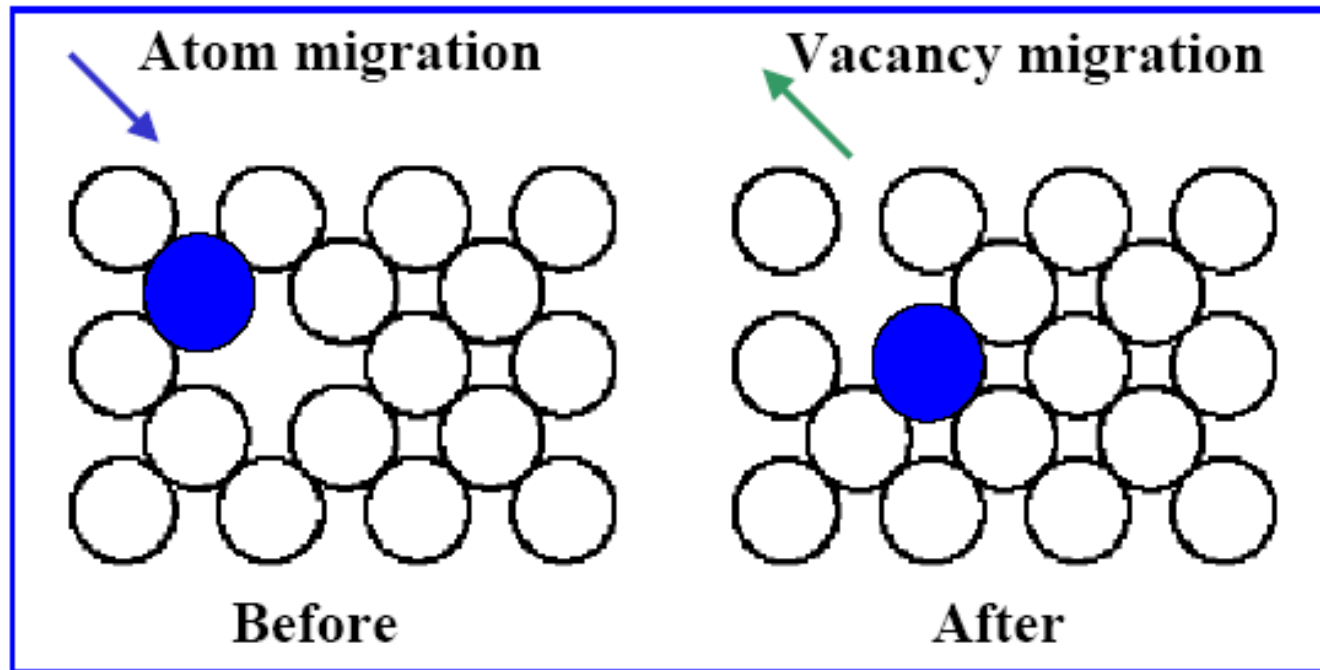


More examples in 3-D !



Diffusion Mechanisms (I)

Vacancy diffusion

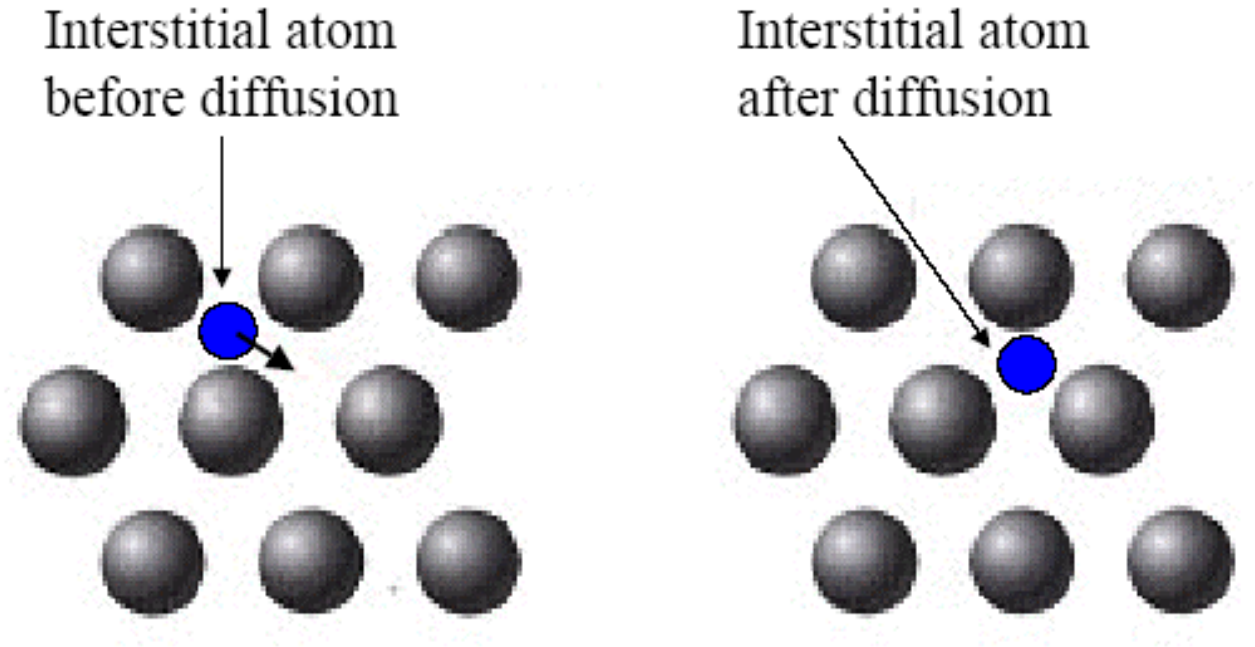


Energy is needed to generate a vacancy, break bonds, cause distortions. Provided by HEAT , kT !

Atom moves in the opposite direction of the vacancy !

Diffusion Mechanisms (II)

Interstitial Diffusion

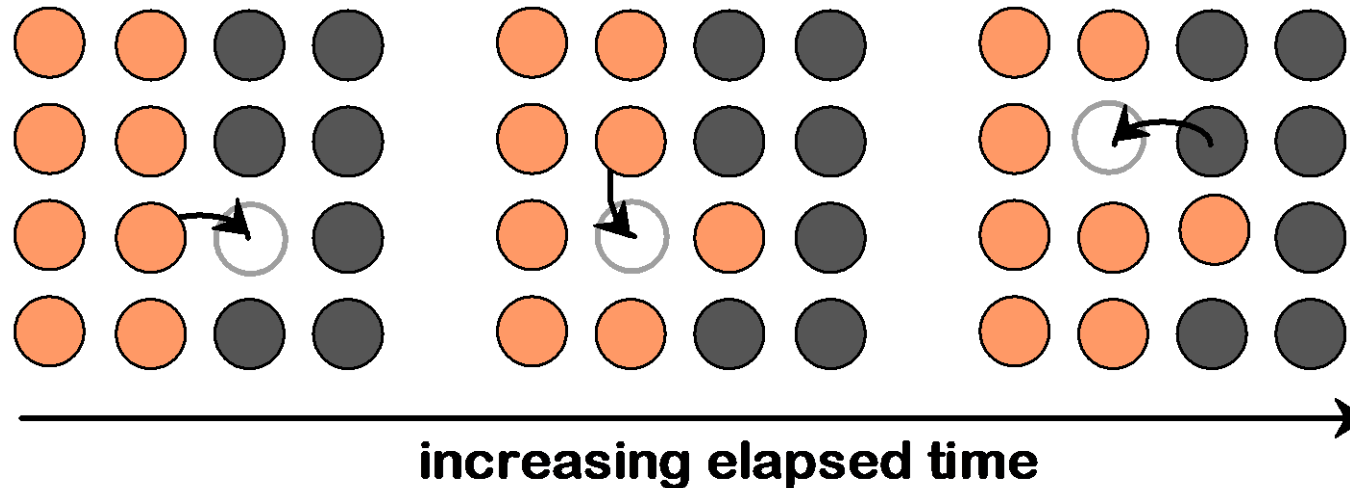


Much faster than vacancy diffusion, why ? Smaller atoms like B, C, H, O. Weaker interaction with the larger atoms. More vacant sites, no need to create a vacancy !

Diffusion Mechanisms (III)

Substitutional Diffusion:

- applies to substitutional impurities
- atoms exchange with vacancies
- rate depends on:
 - number of vacancies
 - activation energy to exchange.

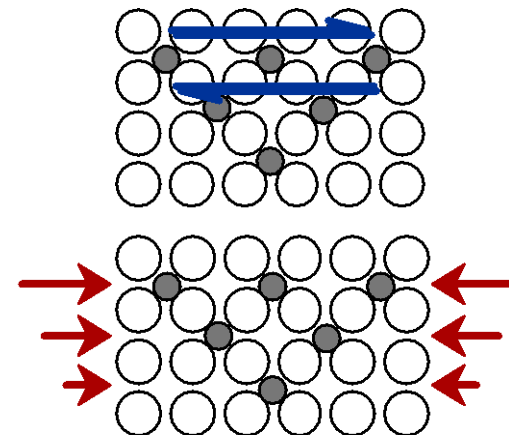


PROCESSING USING DIFFUSION (1)

- **Case Hardening:**
 - Diffuse carbon atoms into the host iron atoms at the surface.
 - Example of interstitial diffusion is a case hardened gear.
- **Result: The "Case" is**
 - hard to deform: C atoms "lock" planes from **shearing**.
 - hard to crack: C atoms put the surface in **compression**.



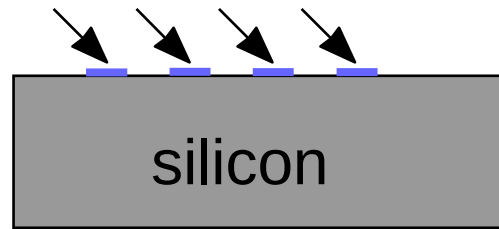
Fig. 5.0,
Callister 6e.
(Fig. 5.0 is
courtesy of
Surface
Division,
Midland-
Ross.)



Processing Using Diffusion

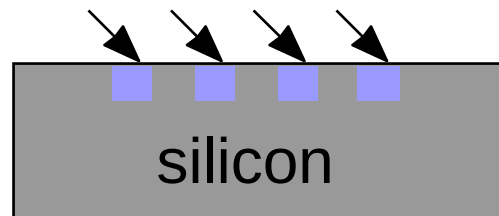
- **Doping** silicon with phosphorus for n -type semiconductors:
- Process:

1. Deposit **P** rich layers on surface.

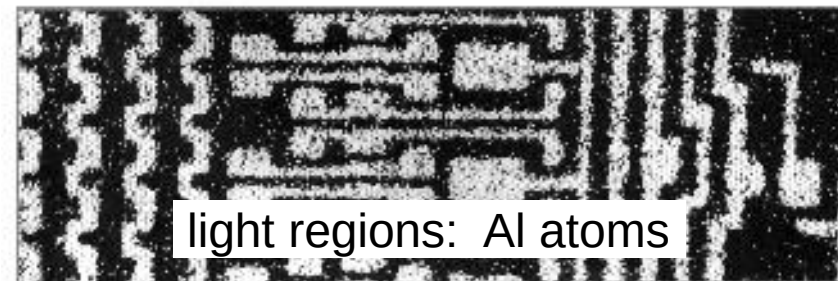
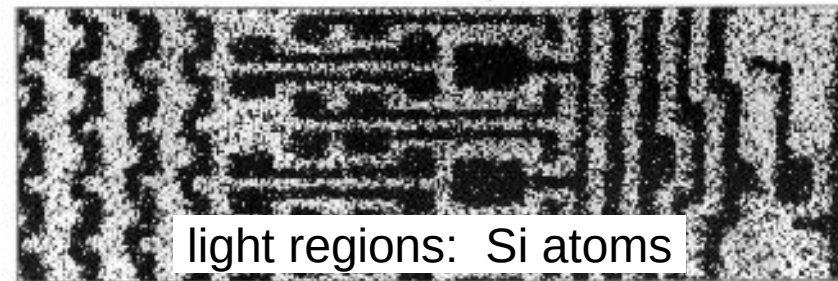
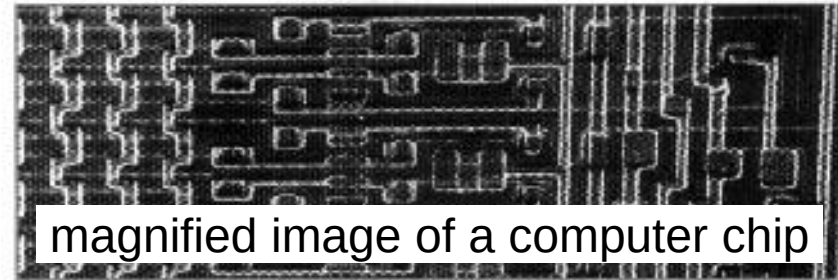


2. Heat it.

3. Result: Doped semiconductor regions.



← 0.5 mm →

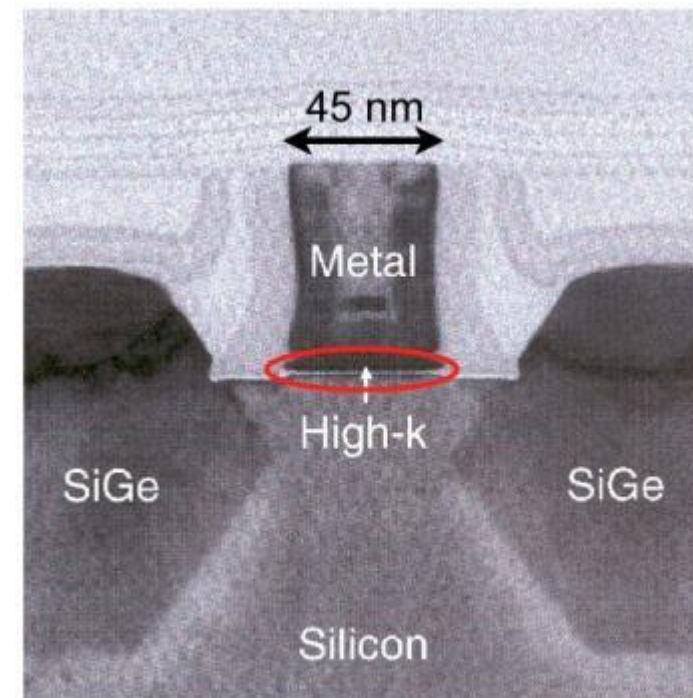


Adapted from chapter-opening photograph,
Chapter 18, *Callister 7e*.

Example – MOSFET

- MOSFET
 - Metal-Oxide-Semiconductor
 - Field-effect transistor
 - A simple switch
 - ~ 6 billion FET in each state-of-the-art transistor

TEM image of a single transistor from Intel's 45nm technology



Adapted from:
<https://volga.eng.yale.edu/teaching-resources/semiconductors>

Diffusion

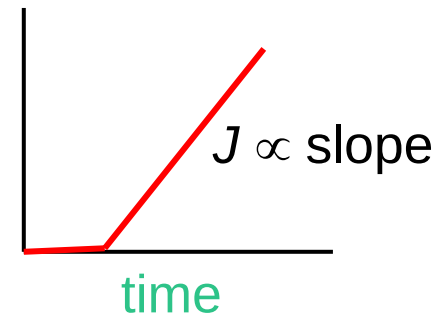
- How do we quantify the amount or rate of diffusion?

$$J \equiv \text{Flux} \equiv \frac{\text{moles (or mass) diffusing}}{(\text{surface area})(\text{time})} = \frac{\text{mol}}{\text{cm}^2\text{s}} \text{ or } \frac{\text{kg}}{\text{m}^2\text{s}}$$

- Measured empirically
 - Make thin film (membrane) of known surface area
 - Impose concentration gradient
 - Measure how fast atoms or molecules diffuse through the membrane

$$J = \frac{M}{At} = \frac{l}{A} \frac{dM}{dt}$$

M =
mass
diffused



MODELING DIFFUSION: FLUX

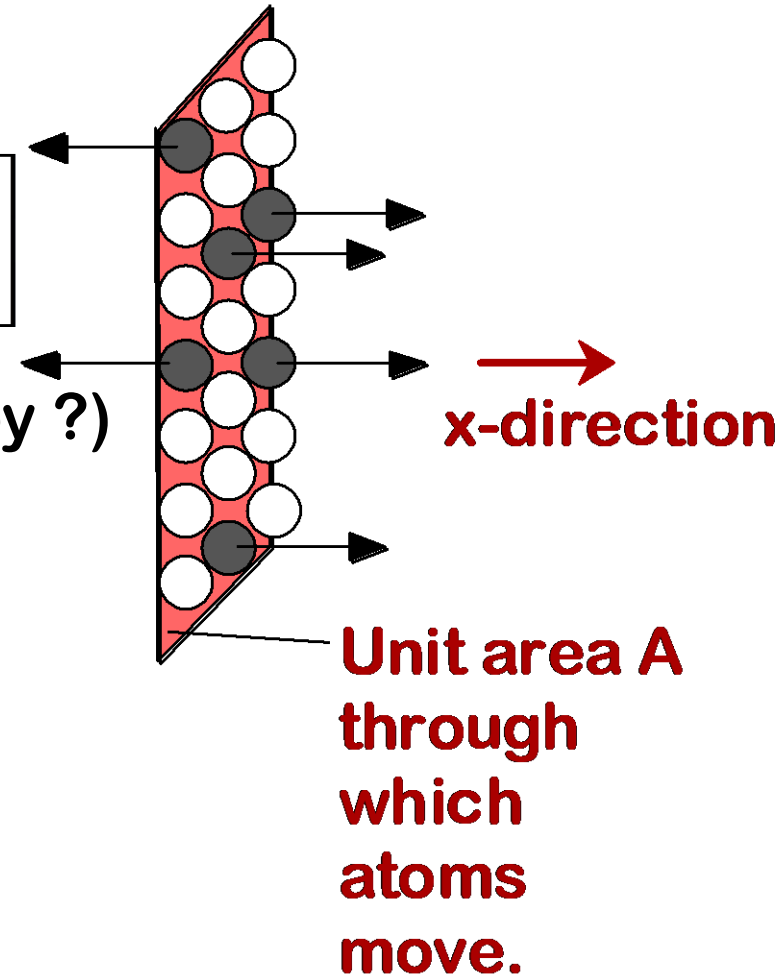
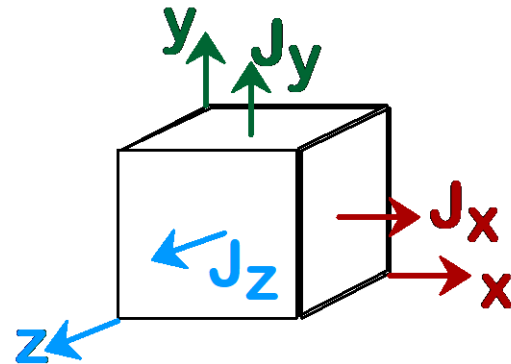
RATE OF MATERIAL TRANSPORT

- **Diffusion**

Flux:
$$J = \frac{1}{A} \frac{dM}{dt} \Rightarrow \left[\frac{\text{kg}}{\text{m}^2\text{s}} \right] \text{ or } \left[\frac{\text{atoms}}{\text{m}^2\text{s}} \right]$$

Material

- Directional Quantity (anisotropy ?)

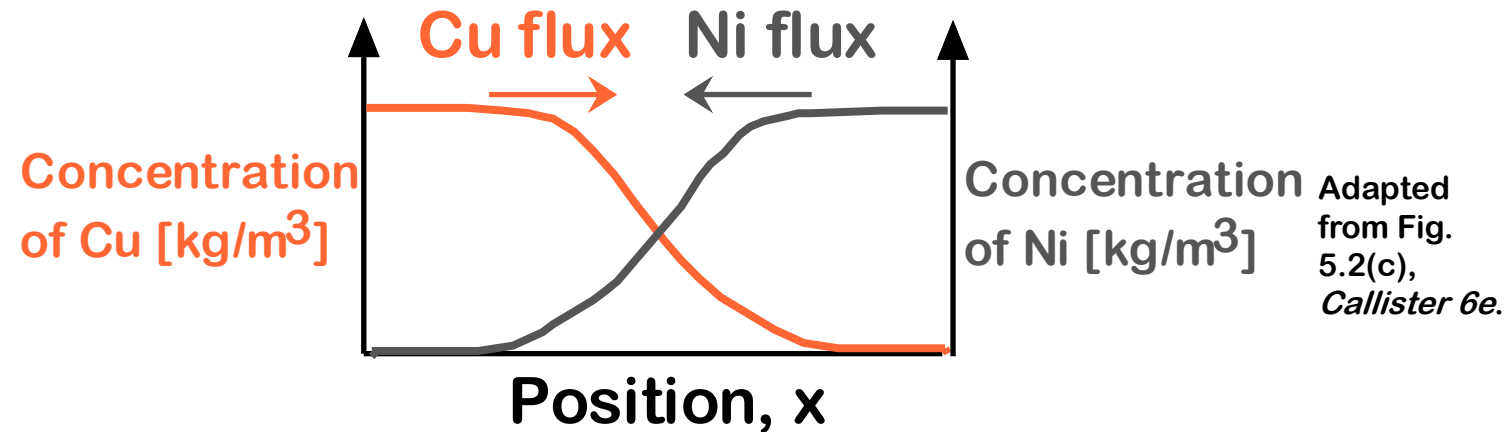


- Flux can be measured for:
 - vacancies
 - host (A) atoms
 - impurity (B) atoms

Diffusion is a time-dependent process !

CONCENTRATION PROFILES & FLUX

- **Concentration Profile, $C(x)$: [kg/m³]**



- **Fick's First Law:**

flux in x-dir.
[kg/m²-s] $\rightarrow J_x = -D \frac{dC}{dx}$

Diffusion coefficient [m²/s] $\rightarrow D$

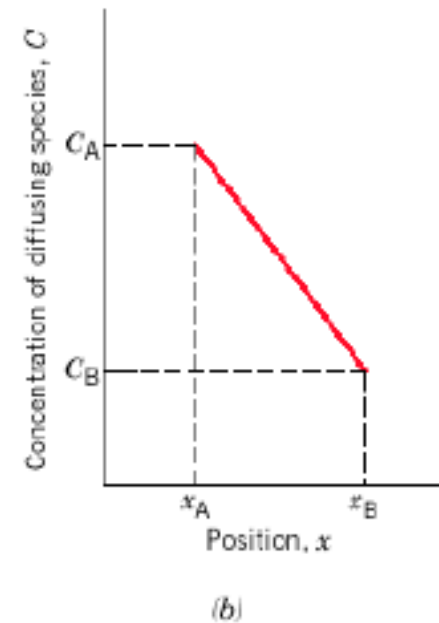
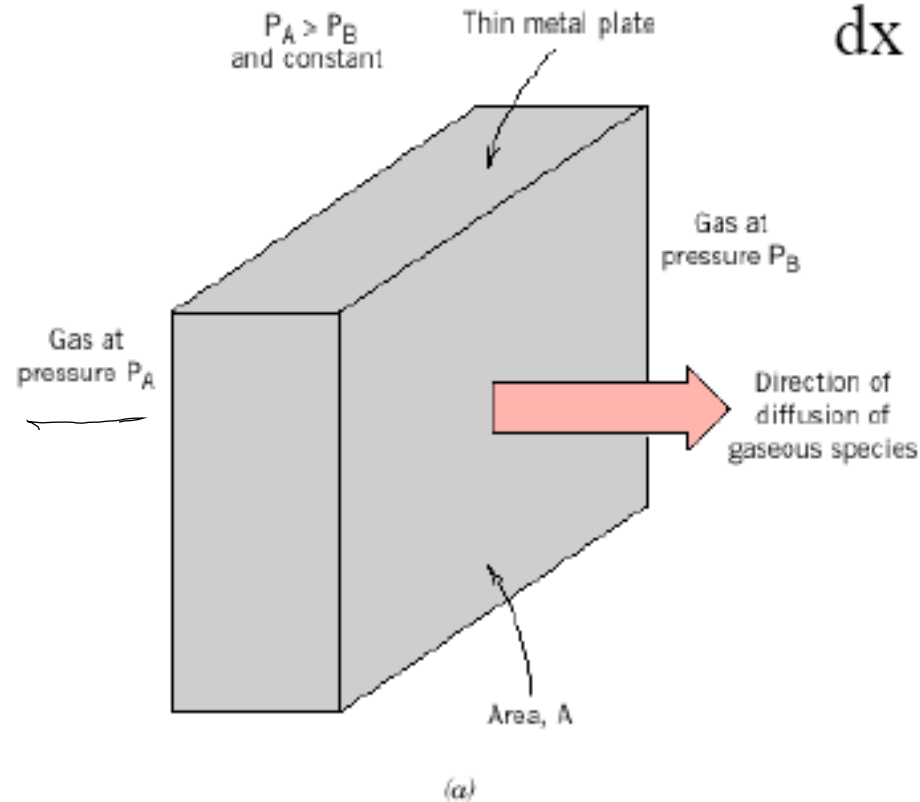
concentration gradient [kg/m⁴] $\rightarrow \frac{dC}{dx}$

- The steeper the concentration profile, the greater the flux!
Concentration gradient is the **DRIVING FORCE** !

Concentration Gradient

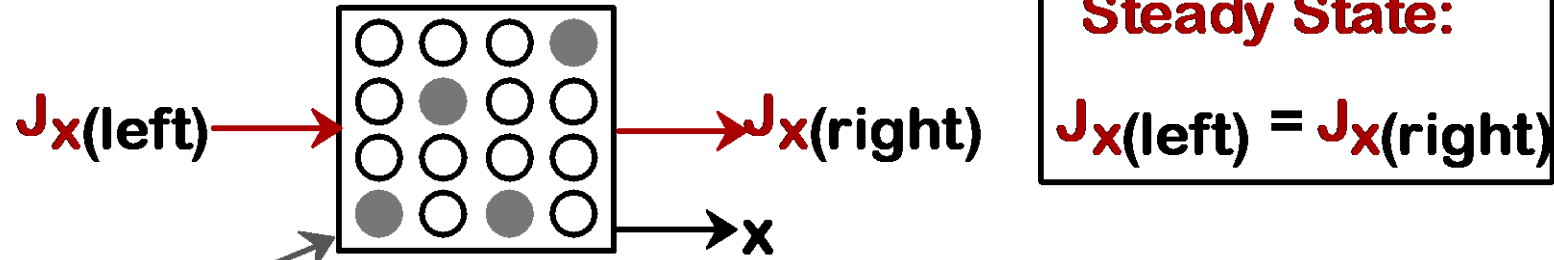
Concentration gradient: dC/dx (Kg.m^{-3}): the slope at a particular point on concentration profile.

$$\frac{dC}{dx} \cong \frac{\Delta C}{\Delta x} = \frac{C_A - C_B}{x_A - x_B}$$



STEADY STATE DIFFUSION

- **Steady State:** the concentration profile doesn't change with time.

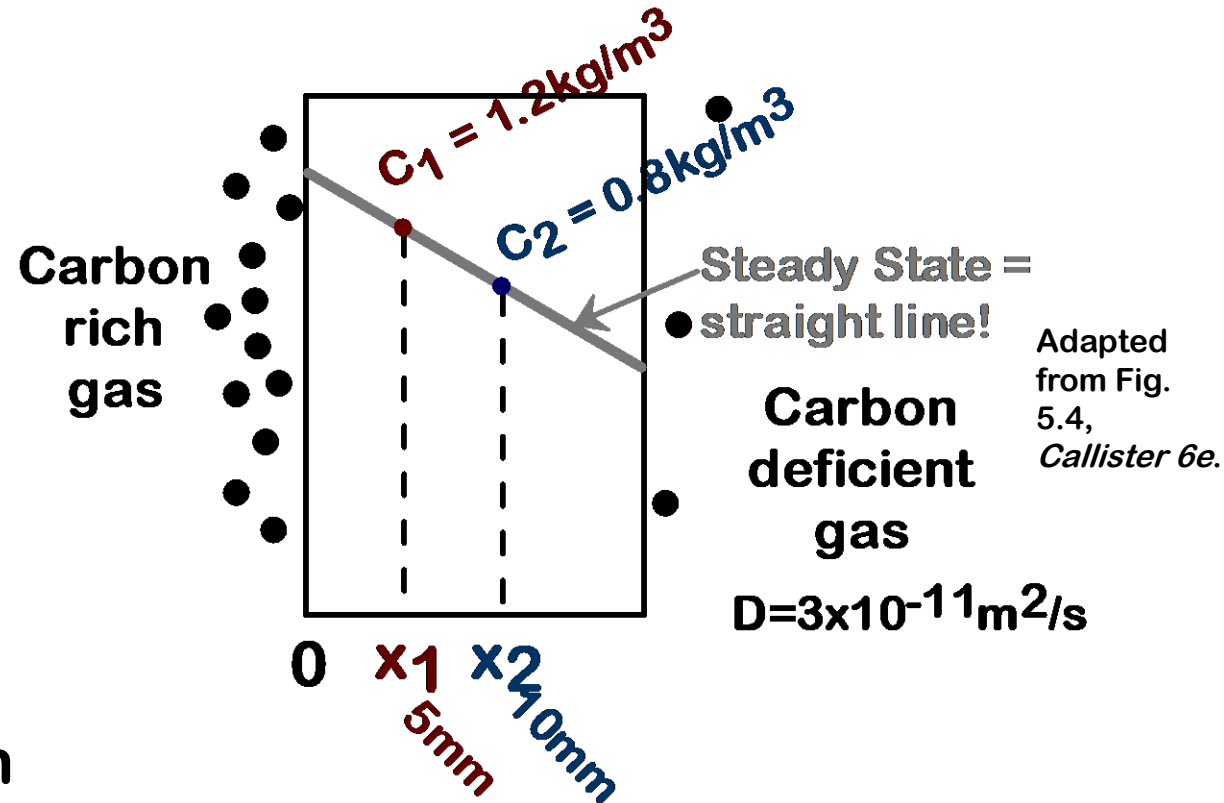


Concentration, C , in the box doesn't change w/time.

- Apply Fick's First Law: $J_x = -D \frac{dC}{dx}$ Why is the minus sign ?
- If $J_x)_{\text{left}} = J_x)_{\text{right}}$, then $\left(\frac{dC}{dx} \right)_{\text{left}} = \left(\frac{dC}{dx} \right)_{\text{right}}$
- Result: the slope, dC/dx , must be constant (i.e., slope doesn't vary with position)!

EX: STEADY STATE DIFFUSION

- Steel plate at 700° C



- Q: How much carbon transfers from the rich to the deficient side?

$$J = -D \frac{C_2 - C_1}{x_2 - x_1} = 2.4 \times 10^{-9} \frac{\text{kg}}{\text{m}^2\text{s}}$$

Example: Chemical Protective Clothing (CPC)

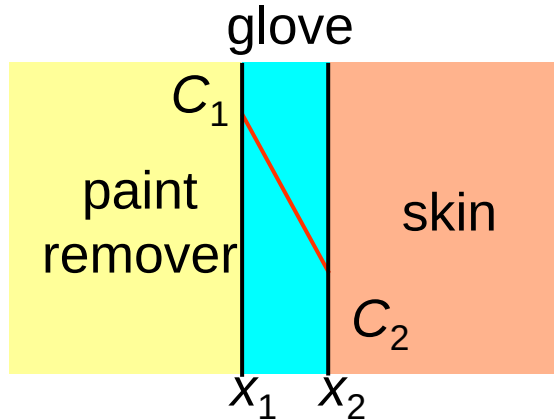
- Methylene chloride is a common ingredient of paint removers. Besides being an irritant, it also may be absorbed through skin. When using this paint remover, protective gloves should be worn.
- If butyl rubber gloves (0.04 cm thick) are used, what is the diffusive flux of methylene chloride through the glove?
- Data:
 - diffusion coefficient in butyl rubber:
 $D = 110 \times 10^{-8} \text{ cm}^2/\text{s}$
 - surface concentrations:

$$C_1 = 0.44 \text{ g/cm}^3$$

$$C_2 = 0.02 \text{ g/cm}^3$$

Example (cont).

- **Solution** – assuming linear conc. gradient



$$J = -D \frac{dC}{dx} \cong -D \frac{C_2 - C_1}{x_2 - x_1}$$

Data: $D = 110 \times 10^{-8} \text{ cm}^2/\text{s}$

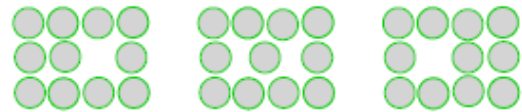
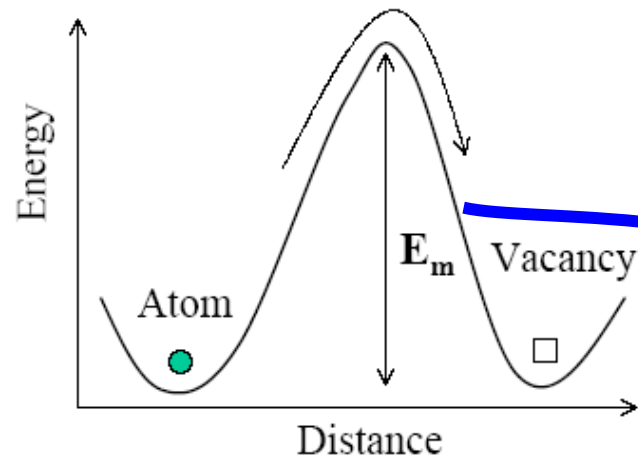
$$C_1 = 0.44 \text{ g/cm}^3$$

$$C_2 = 0.02 \text{ g/cm}^3$$

$$x_2 - x_1 = 0.04 \text{ cm}$$

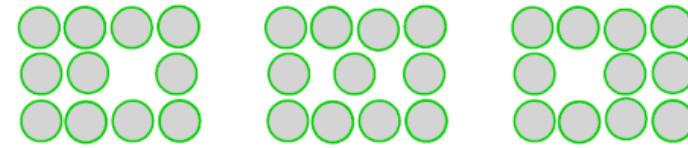
$$J = - (110 \times 10^{-8} \text{ cm}^2/\text{s}) \frac{(0.02 \text{ g/cm}^3 - 0.44 \text{ g/cm}^3)}{(0.04 \text{ cm})} = 1.16 \times 10^{-5} \frac{\text{g}}{\text{cm}^2\text{s}}$$

Temperature Dependency !



Atom has to break bonds and “squeeze” thru => activation energy, $E_m \approx 1 \text{ eV}$.

$$R_j = R_0 \exp\left(-\frac{E_m}{k_B T}\right)$$



What is the probability to find a vacancy at a nearest site ?

$$P = C.N. \exp\left(-\frac{Q_v}{k_B T}\right)$$

COMBINE

Temperature Effect!

$$D \approx C.N.R_0 a^2 \exp\left(-\frac{E_m}{k_B T}\right) \exp\left(-\frac{Q_v}{k_B T}\right) =$$
$$= D_0 \exp\left(-\frac{(E_m + Q_v)}{k_B T}\right) = \underline{D_0 \exp\left(-\frac{Q_d}{k_B T}\right)}$$

The diffusion depends on temperature because;

- a- # of vacancies in the vicinity
- b- thermally activated successful jumps

Diffusion and Temperature

- Diffusion coefficient increases with increasing T .

$$D = D_o \exp\left(-\frac{Q_d}{RT}\right)$$

D = diffusion coefficient [m^2/s]

D_o = pre-exponential [m^2/s]

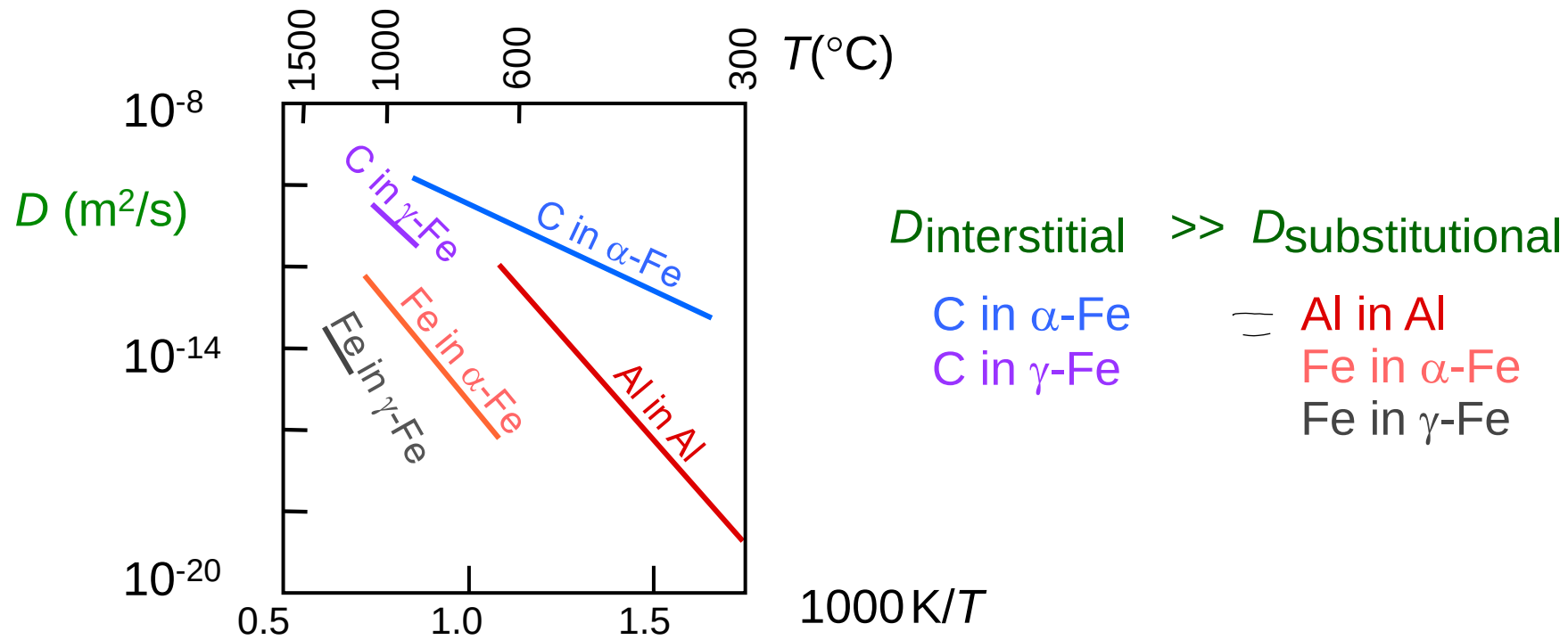
Q_d = activation energy [J/mol or eV/atom]

R = gas constant [8.314 J/mol-K]

T = absolute temperature [K]

Diffusion and Temperature

D has exponential dependence on T



Adapted from Fig. 5.7, *Callister 7e*. (Date for Fig. 5.7 taken from E.A. Brandes and G.B. Brook (Ed.) *Smithells Metals Reference Book*, 7th ed., Butterworth-Heinemann, Oxford, 1992.)

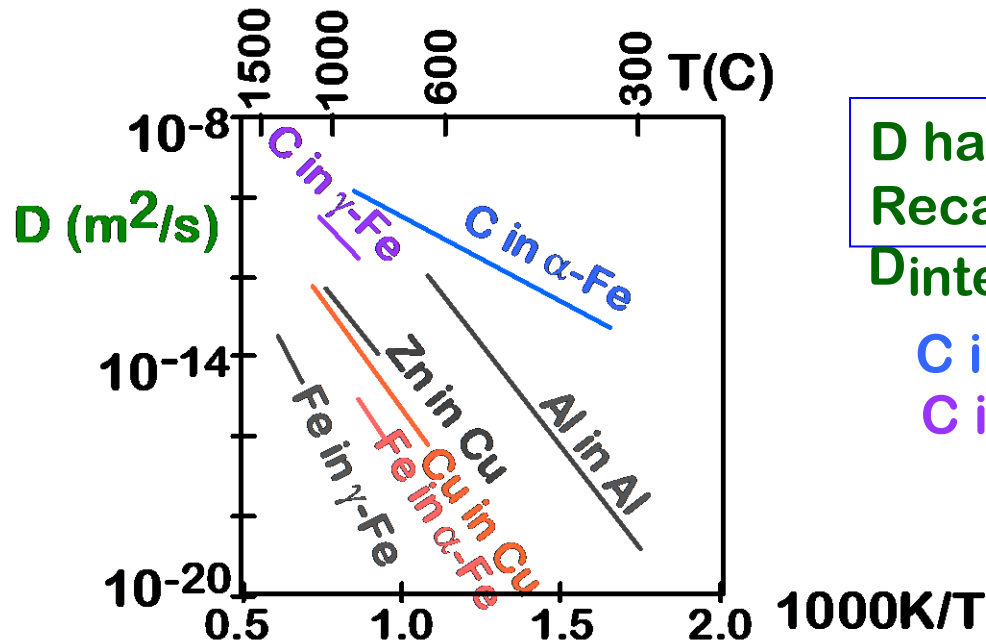
DIFFUSION AND TEMPERATURE

- Diffusivity increases with T.

diffusivity $D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$

D_0 : pre-exponential [m^2/s] (see Table 5.2, *Callister 6e*)
 Q_d : activation energy [J/mol], [eV/mol] (see Table 5.2, *Callister 6e*)
 R : gas constant [8.31 J/mol-K]

- Experimental Data:



D has exp. dependence on T
Recall: Vacancy does also!

$D_{\text{interstitial}} \gg D_{\text{substitutional}}$

C in α -Fe
 C in γ -Fe

Cu in Cu
 Al in Al
 Fe in α -Fe
 Fe in γ -Fe
 Zn in Cu

Adapted from Fig. 5.7, *Callister 6e*. (Date for Fig. 5.7 taken from E.A. Brandes and G.B. Brook (Ed.) *Smithells Metals Reference Book*, 7th ed., Butterworth-Heinemann, Oxford, 1992.)

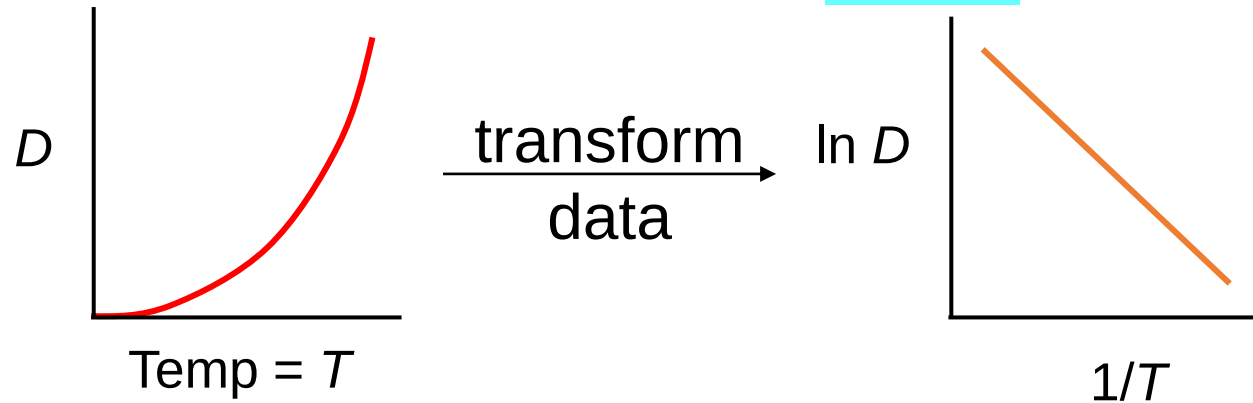
Example: At 300°C the diffusion coefficient and activation energy for Cu in Si are

$$D(300^\circ\text{C}) = 7.8 \times 10^{-11} \text{ m}^2/\text{s}$$

$$Q_d = 41.5 \text{ kJ/mol}$$

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

What is the diffusion coefficient at 350°C?



$$\ln D_2 = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T_2} \right) \quad \text{and} \quad \ln D_1 = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T_1} \right)$$

$$\therefore \ln D_2 - \ln D_1 = \ln \frac{D_2}{D_1} = -\frac{Q_d}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

Example (cont.)

$$D_2 = D_1 \exp \left[-\frac{Q_d}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \right]$$

$$T_1 = 273 + 300 = 573 \text{ K}$$

$$T_2 = 273 + 350 = 623 \text{ K}$$

$$D_2 = (7.8 \times 10^{-11} \text{ m}^2/\text{s}) \exp \left[\frac{-41,500 \text{ J/mol}}{8.314 \text{ J/mol} \cdot \text{K}} \left(\frac{1}{623 \text{ K}} - \frac{1}{573 \text{ K}} \right) \right]$$

$$D_2 = 15.7 \times 10^{-11} \text{ m}^2/\text{s}$$

Fick's Second Law ; Non-steady state Diffusion

- In most practical cases, J (flux) and $\frac{dC}{dx}$ (concentration gradient) change with time (t).
 - Net accumulation or depletion of species diffusing
- How do we express a time dependent concentration?

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right) = D \frac{\partial^2 C}{\partial x^2}$$

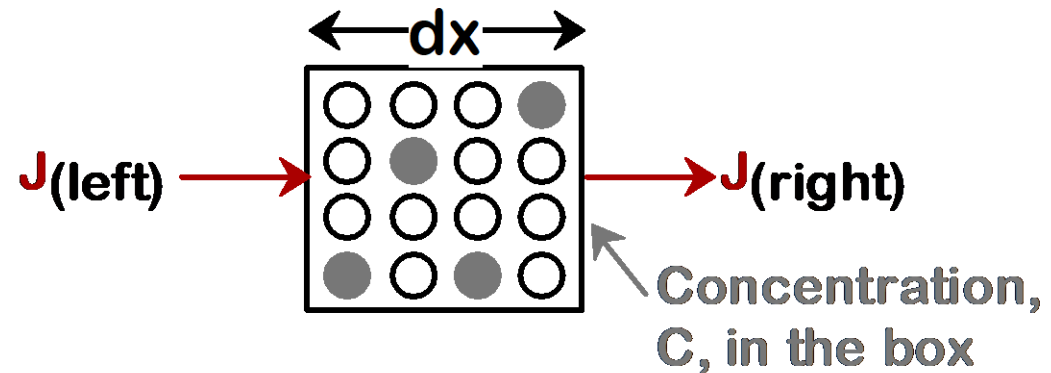
Concentration at a point x
Changing with time

?

Flux, J , changes
at any point x !

NON STEADY STATE DIFFUSION

- Concentration profile, $C(x)$, changes w/ time.



- To conserve matter:

$$\frac{J(\text{right}) - J(\text{left})}{dx} = - \frac{dC}{dt}$$

$$\frac{dJ}{dx} = - \frac{dC}{dt}$$

- Fick's First Law:

$$J = -D \frac{dC}{dx} \quad \text{or}$$

$$\frac{dJ}{dx} = -D \frac{d^2C}{dx^2} \quad (\text{if } D \text{ does not vary with } x)$$

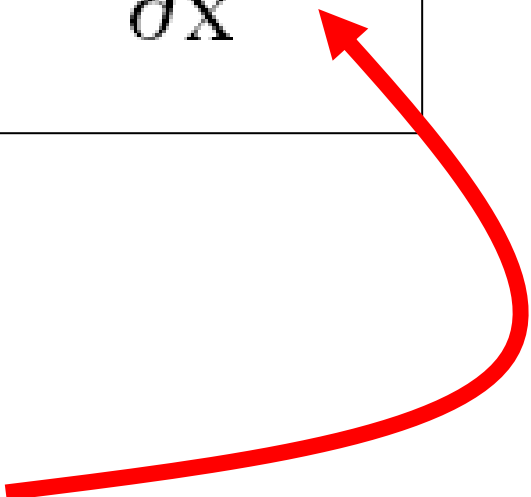
equate

- Governing Eqn.:

$$\frac{dC}{dt} = D \frac{d^2C}{dx^2}$$

How do we solve this partial differential equation ?

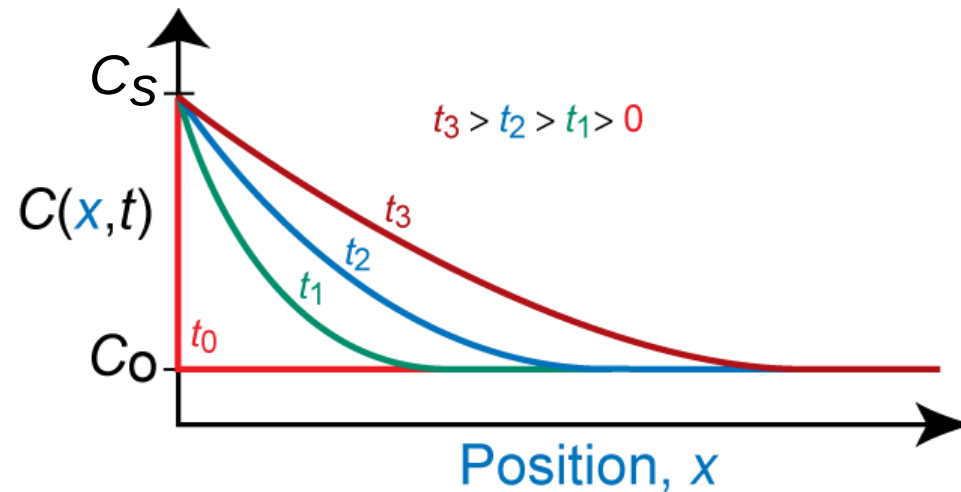
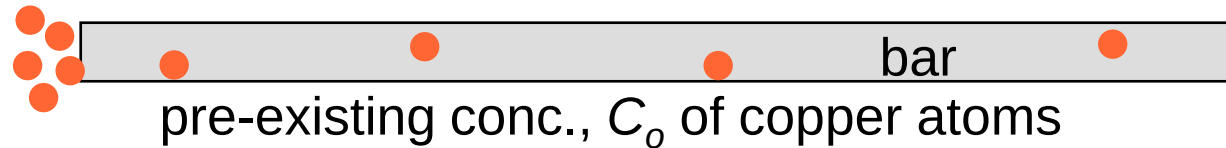
$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right) = D \frac{\partial^2 C}{\partial x^2}$$

- Use proper boundary conditions:
 - $t=0, C = C_0$, at $0 \leq x \leq \infty$
 - $t>0, C = C_s$, at $x = 0$
 $C = C_0$, at $x = \infty$
- 

Non-steady State Diffusion

- Copper diffuses into a bar of aluminum.

Surface conc.,
 C_S of Cu atoms



Adapted from
Fig. 5.5,
Callister 7e.

B.C. at $t = 0$, $C = C_0$ for $0 \leq x \leq \infty$

at $t > 0$, $C = C_S$ for $x = 0$ (const. surf. conc.)

$C = C_0$ for $x = \infty$

Solution:

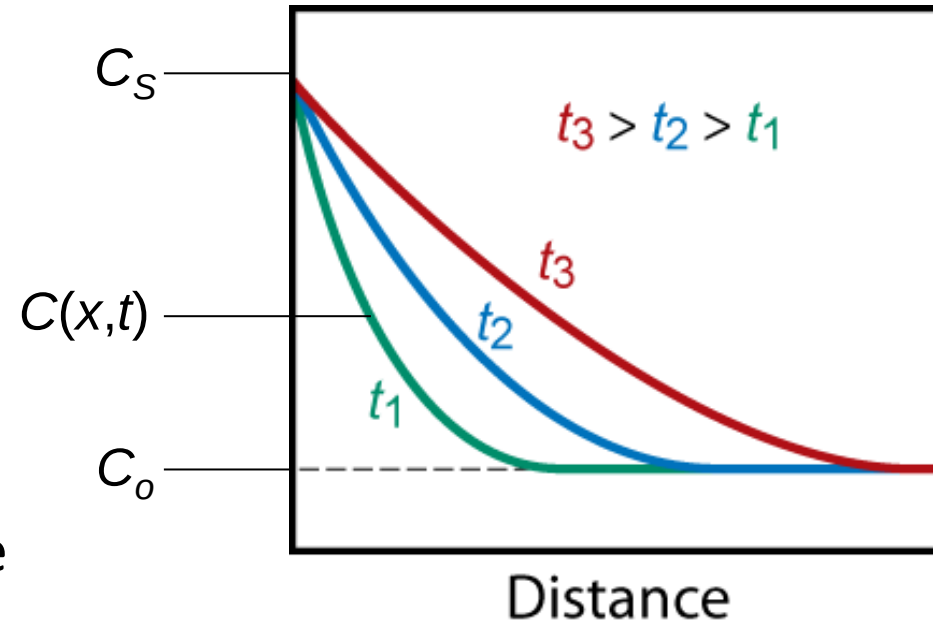
$$\frac{C(x,t) - C_o}{C_s - C_o} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$C(x,t)$ = Conc. at point x at time t

$\operatorname{erf}(z)$ = error function

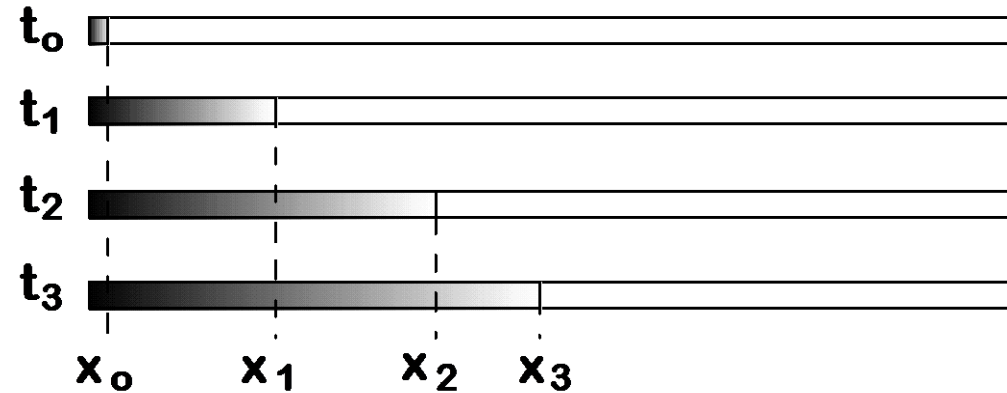
$$= \frac{2}{\sqrt{\pi}} \int_0^z e^{-y^2} dy$$

$\operatorname{erf}(z)$ values are given in Table 5.1



DIFFUSION DEMO: ANALYSIS

- The experiment: we recorded combinations of t and x that kept C constant.



$$\frac{C(x_i, t_i) - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x_i}{2\sqrt{Dt_i}}\right) = \text{(constant here)}$$

- Diffusion depth given by:

$$x_i \propto \sqrt{Dt_i}$$

Non-steady State Diffusion

- Sample Problem: An FCC iron-carbon alloy initially containing 0.20 wt% C is carburized at an elevated temperature and in an atmosphere that gives a surface carbon concentration constant at 1.0 wt%. If after 49.5 h the concentration of carbon is 0.35 wt% at a position 4.0 mm below the surface, determine the temperature at which the treatment was carried out.
- **Solution:** use Eqn. 5.5

$$\frac{C(x,t) - C_o}{C_s - C_o} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Solution (cont.):

$$\frac{C(x,t) - C_o}{C_s - C_o} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$



- $t = 49.5 \text{ h}$
 - $C_x = 0.35 \text{ wt\%}$
 - $C_o = 0.20 \text{ wt\%}$
- $x = 4 \times 10^{-3} \text{ m}$
 $C_s = 1.0 \text{ wt\%}$

$$\frac{C(x,t) - C_o}{C_s - C_o} = \frac{0.35 - 0.20}{1.0 - 0.20} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) = 1 - \operatorname{erf}(z)$$

$$\therefore \operatorname{erf}(z) = 0.8125$$

Solution (cont.):

We must now determine from Table 5.1 the value of z for which the error function is 0.8125. An interpolation is necessary as follows

z	$\text{erf}(z)$
0.90	0.7970
z	0.8125
0.95	0.8209

$$\frac{z - 0.90}{0.95 - 0.90} = \frac{0.8125 - 0.7970}{0.8209 - 0.7970}$$

$$z = 0.93$$

Now solve for D

$$z = \frac{x}{2\sqrt{Dt}} \Rightarrow D = \frac{x^2}{4z^2t}$$

$$\therefore D = \left(\frac{x^2}{4z^2t} \right) = \frac{(4 \times 10^{-3} \text{ m})^2}{(4)(0.93)^2 (49.5 \text{ h})} \frac{1 \text{ h}}{3600 \text{ s}} = 2.6 \times 10^{-11} \text{ m}^2/\text{s}$$

Solution (cont.):

- To solve for the temperature at which D has above value, we use a rearranged form of Equation (5.9a);

$$T = \frac{Q_d}{R(\ln D_o - \ln D)}$$

from Table 5.2, for diffusion of C in FCC Fe

$$D_o = 2.3 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Q_d = 148,000 \text{ J/mol}$$

$$\therefore T = \frac{148,000 \text{ J/mol}}{(8.314 \text{ J/mol} \cdot \text{K})(\ln 2.3 \times 10^{-5} \text{ m}^2/\text{s} - \ln 2.6 \times 10^{-11} \text{ m}^2/\text{s})}$$

$$T = 1300 \text{ K} = 1027^\circ\text{C}$$

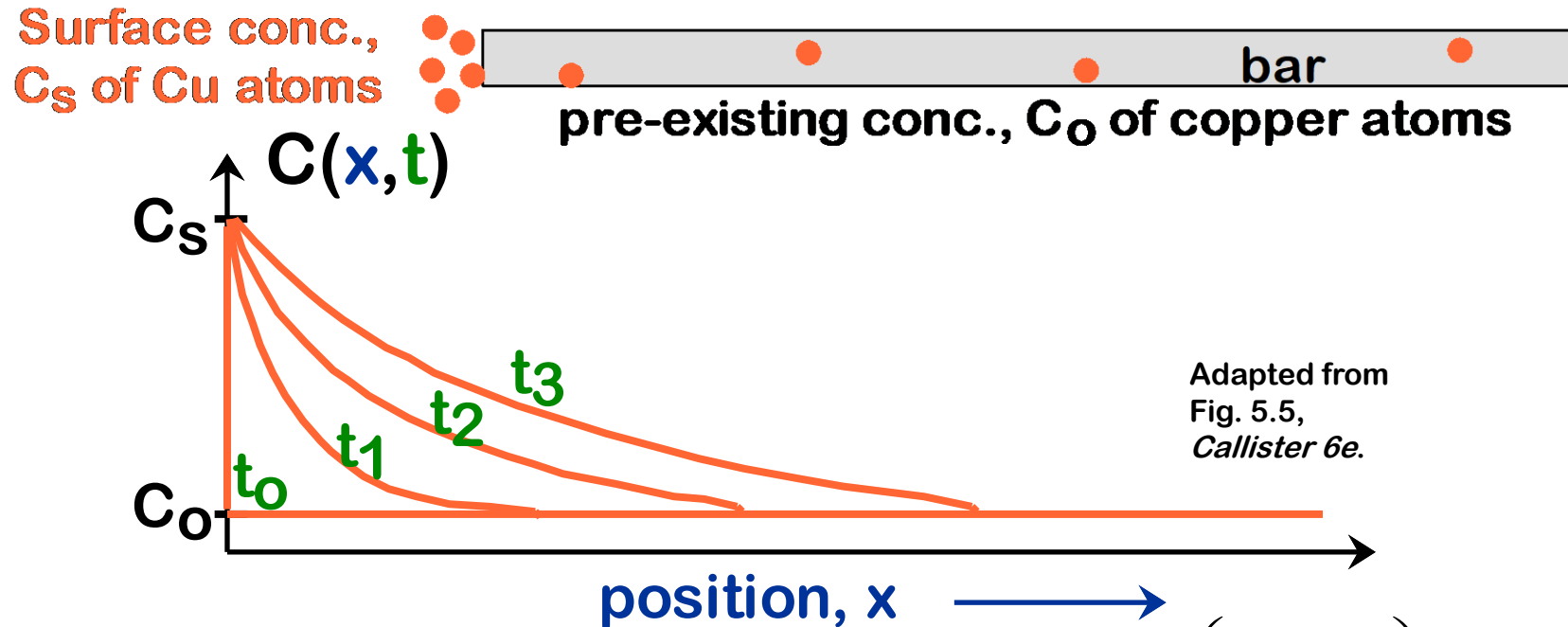
Non-steady State Diffusion

- Sample Problem: We have tube filled with water from a North Carolina shore to Nice in southern France. We drop an ink droplet in NC at $t=0$. How much ink will reach Nice in 1 second?

- **Solution:** use Eqn. 5.5
$$\frac{C(x,t) - C_o}{C_s - C_o} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

EX: NON STEADY STATE DIFFUSION

- Copper diffuses into a bar of aluminum.

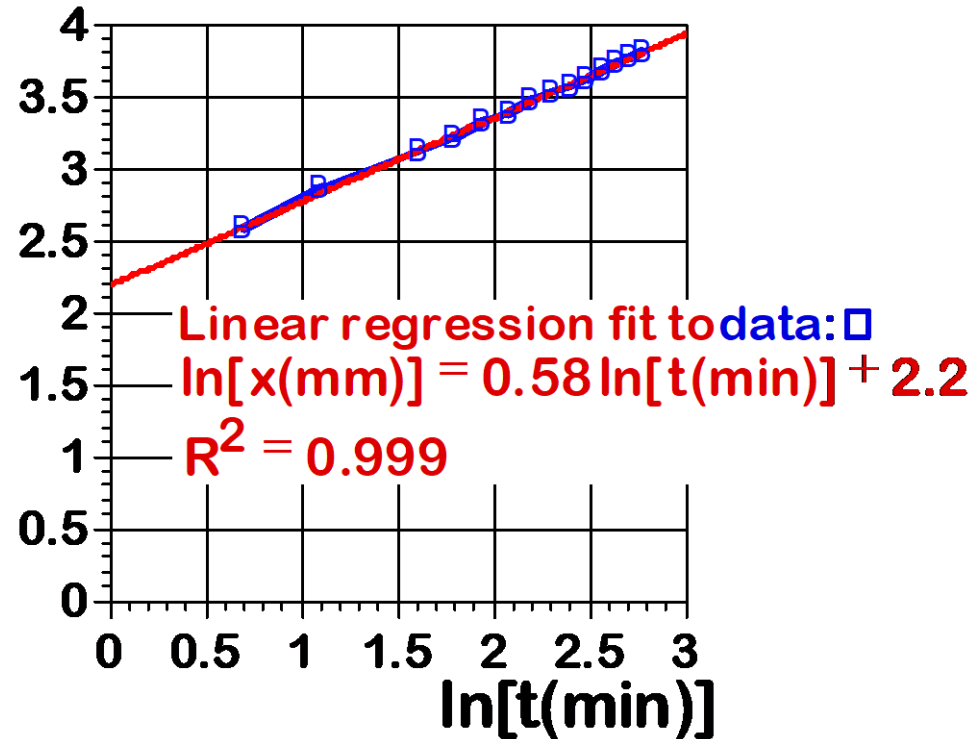


- General solution:
$$\frac{C(x, t) - C_o}{C_s - C_o} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$C(x, t)$ = concentration at any time and position !

"Gaussian error function"

DATA FROM DIFFUSION DEMO



- Experimental result: $x \sim t^{0.58}$
- Theory predicts $x \sim t^{0.50}$
- Reasonable agreement!

Example 5.3

- Copper diffuses into a bar of aluminum.
- 10 hours at 600C gives desired C(x).
- How many hours would it take to get the same C(x) if we processed at 500C?

Key point 1: $C(x, t_{500C}) = C(x, t_{600C})$.

Key point 2: Both cases have the same C_0 and C_s .

- Result: Dt should be held constant.

$$\frac{C(x, t) - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{\sqrt{2Dt}}\right) \rightarrow (Dt)_{500^\circ\text{C}} = (Dt)_{600^\circ\text{C}}$$

• Answer: $t_{500} = \frac{(Dt)_{600}}{D_{500}} = 110\text{hr}$

$5.3 \times 10^{-13} \text{m}^2/\text{s}$ points to $(Dt)_{600}$
 10hrs points to $(Dt)_{600}$
 $4.8 \times 10^{-14} \text{m}^2/\text{s}$ points to D_{500}

Note: values of D are provided here.

Size Impact on Diffusion

Diffusion of different species

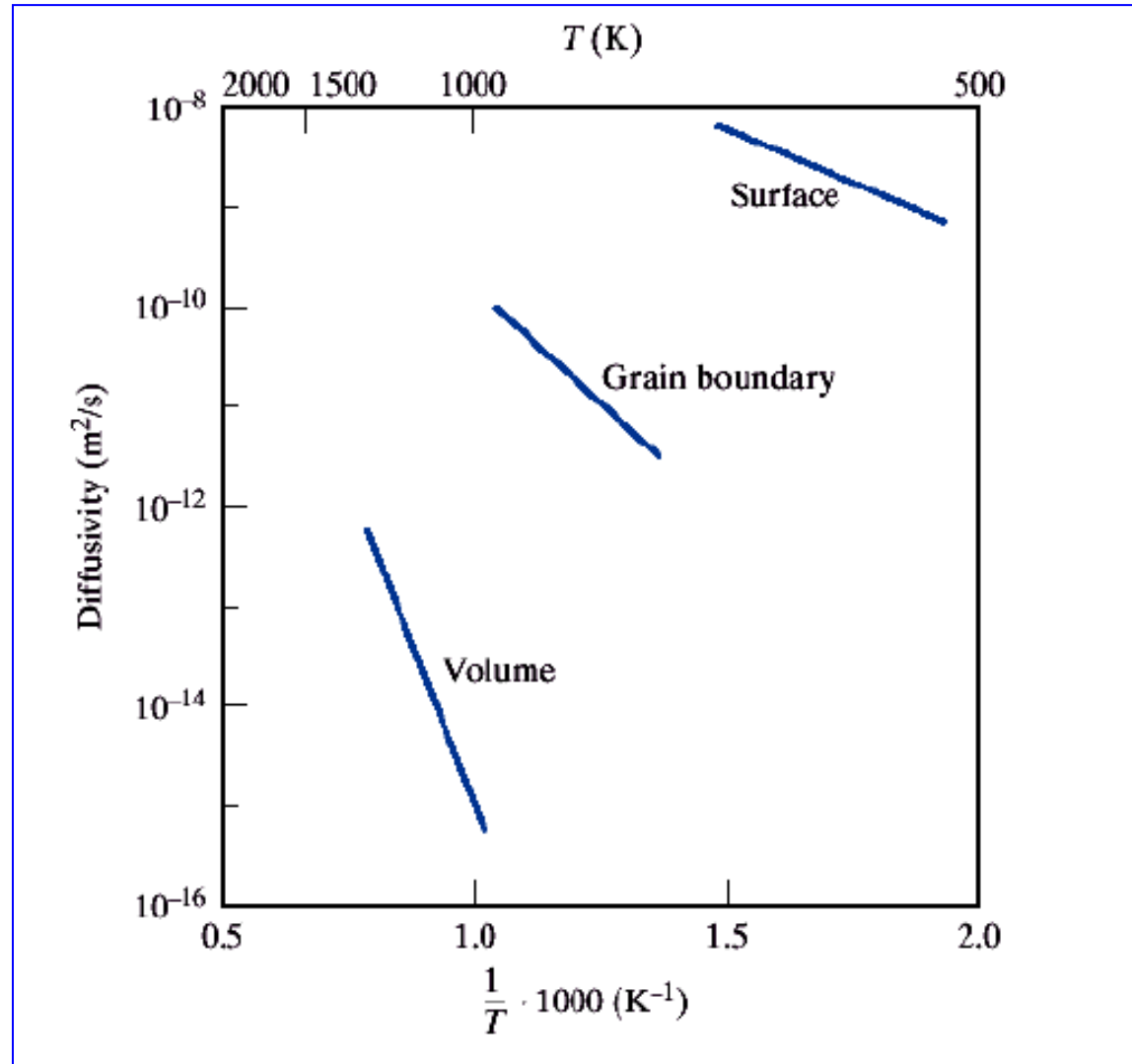
Table 5.2 A Tabulation of Diffusion Data

Diffusing Species	Host Metal	$D_0(\text{m}^2/\text{s})$	Activation Energy Q_d		Calculated Values	
			kJ/mol	eV/atom	$T(^{\circ}\text{C})$	$D(\text{m}^2/\text{s})$
Fe	α -Fe (BCC)	2.8×10^{-5}	251	2.60	500	3.0×10^{-21}
					900	1.8×10^{-15}
Fe	γ -Fe (FCC)	5.0×10^{-6}	284	2.94	900	1.1×10^{-17}
					1100	7.8×10^{-16}
C	α -Fe	6.2×10^{-6}	80	0.83	500	2.4×10^{-12}
					900	1.7×10^{-10}
C	γ -Fe	2.3×10^{-6}	148	1.53	900	5.9×10^{-12}
					1100	5.3×10^{-11}
Cu	Cu	7.8×10^{-6}	211	2.19	500	4.2×10^{-19}
Zn	Cu	2.4×10^{-6}	189	1.96	500	4.0×10^{-18}
Al	Al	2.3×10^{-6}	144	1.49	500	4.2×10^{-14}
Cu	Al	6.5×10^{-6}	136	1.41	500	4.1×10^{-14}
Mg	Al	1.2×10^{-6}	131	1.35	500	1.9×10^{-13}
Cu	Ni	2.7×10^{-6}	256	2.65	500	1.3×10^{-22}

Source: E. A. Brandes and G. B. Brook (Editors), *Smithells Metals Reference Book*, 7th edition, Butterworth-Heinemann, Oxford, 1992.

Smaller atoms diffuse faster

Fast Tracks for diffusion !



e.g. self-diffusion of Ag :
-Areas where lattice is pre-strained can allow for faster diffusion of atoms
-Less energy is needed to distort an already strained lattice !

Important

- **Temperature** - diffusion rate increases with increasing temperature (WHY ?)
- **Diffusion mechanism** – interstitials diffuse faster (WHY ?)
- **Diffusing and host species** - D_0 , Q_d is different for every solute - solvent pair
- **Microstructure** - grain boundaries and dislocation cores provide faster pathways for diffusing species, hence diffusion is faster in polycrystalline vs. single crystal materials. (WHY ?)

SUMMARY:

STRUCTURE & DIFFUSION

Diffusion **FASTER** for...

- open crystal structures
- lower melting T materials
- materials w/secondary bonding
- smaller diffusing atoms
- cations
- lower density materials

WHY ?

Diffusion **SLOWER** for...

- close-packed structures
- higher melting T materials
- materials w/covalent bonding
- larger diffusing atoms
- anions
- higher density materials

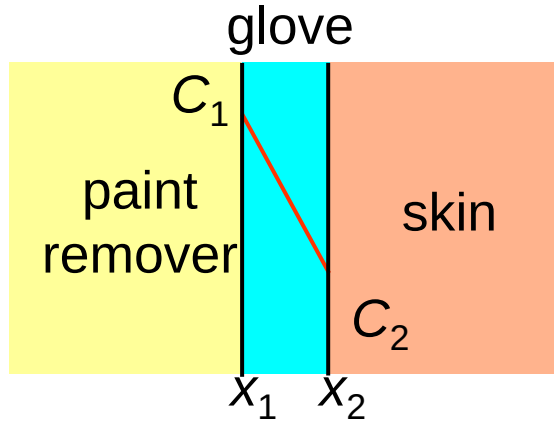
Example: Chemical Protective Clothing (CPC)

- Methylene chloride is a common ingredient of paint removers. Besides being an irritant, it also may be absorbed through skin. When using this paint remover, protective gloves should be worn.
- If butyl rubber gloves (0.04 cm thick) are used, what is the breakthrough time (t_b), i.e., how long could the gloves be used before methylene chloride reaches the hand?
- Data (from Table 22.5)
 - diffusion coefficient in butyl rubber:

$$D = 110 \times 10^{-8} \text{ cm}^2/\text{s}$$

Example (cont).

- **Solution** – assuming linear conc. gradient



$$t_b = \frac{\ell^2}{6D}$$

$$\ell = x_2 - x_1 = 0.04 \text{ cm}$$

$$D = 110 \times 10^{-8} \text{ cm}^2/\text{s}$$

$$t_b = \frac{(0.04 \text{ cm})^2}{(6)(110 \times 10^{-8} \text{ cm}^2/\text{s})} = 240 \text{ s} = 4 \text{ min}$$

Time required for breakthrough ca. **4 min**